

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

04

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Q1**B****B. 8**

Choice B is correct. The standard form of an equation of a circle in the xy -plane is $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is r . For the circle in the xy -plane with equation $x - 5^2 + y - 3^2 = 16$, it follows that $r^2 = 16$. Taking the square root of both sides of this equation yields $r = 4$ or $r = -4$. Because r represents the length of the radius of the circle and this length must be positive, $r = 4$. Therefore, the radius of the circle is 4. The diameter of a circle is twice the length of the radius of the circle. Thus, 24 yields 8. Therefore, the diameter of the circle is 8.

Choice A is incorrect. This is the radius of the circle.

Choice C is incorrect. This is the square of the radius of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D**

D. $\frac{3}{\sqrt{3}}$

Choice D is correct. The tangent of a nonright angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Using that definition for tangent, in a right triangle with legs that have lengths a and b , the tangent of one acute angle is $\frac{a}{b}$ and the tangent for the other acute angle is $\frac{b}{a}$. It follows that the tangents of the acute angles in a right triangle are reciprocals of each other. Therefore, the tangent of the other acute angle in the given triangle is the reciprocal of $\frac{\sqrt{3}}{3}$ or $\frac{3}{\sqrt{3}}$.

Choice A is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the tangent of the angle described. Choice B is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the reciprocal of the tangent of the angle described. Choice C is incorrect and may result from interpreting the tangent of the other acute angle as equal to the tangent of the angle described.

Q3**A****A.** $-14x + 1,540$

Choice A is correct. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. Since there are two parallel lines intersected by a transversal, all four acute angles have the same measure and all four obtuse angles have the same measure.

Additionally, each acute angle is supplementary to each obtuse angle. It's given that the measure of one of these eight angles is $(7x - 250)^\circ$. It follows that a supplementary angle has measure $(180 - (7x - 250))^\circ$, or $(-7x + 430)^\circ$. It's also given that the sum of the measures of four of the eight angles is k° . It follows that the possible values of k are $4(7x - 250)$; $(7x - 250) + 3(-7x + 430)$; $2(7x - 250) + 2(-7x + 430)$; $3(7x - 250) + (-7x + 430)$; and $4(-7x + 430)$. These values are equivalent to $28x - 1,000$; $-14x + 1,040$; 360 ; $14x - 320$; and $-28x + 1,720$, respectively. It follows that of the given choices, only $-14x + 1,540$ could NOT be equivalent to k , for all values of x .

Choice B is incorrect. This is the sum of three angles with measure $(7x - 250)^\circ$ and one angle with measure $(-7x + 430)^\circ$.

Choice C is incorrect. This is the sum of four angles with measure $(-7x + 430)^\circ$.

Choice D is incorrect. This is the sum of two angles with measure $(7x - 250)^\circ$ and two angles with measure $(-7x + 430)^\circ$.

Q4**8**

The correct answer is 8. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(6, 6)$ is halfway between the points $(6, 2)$ and $(6, 10)$, and is a distance of 4 units from each of those two points. The point $(6, 6)$ is also a distance of 4 units from $(2, 6)$. Because the point $(6, 6)$ is the same distance from all three points shown, it must be the center of the circle. Since that distance is 4, it follows that the radius of the circle is 4. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 8.

Q5

A

A. 5

Choice A is correct. One way to find the radius of the circle is to rewrite the given equation in standard form, $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle and the radius of the circle is r . To do this, divide the original equation, $2x^2 - 6x + 2y^2 + 2y = 45$, by 2 to make the leading coefficients of x^2 and y^2 each equal to 1: $x^2 - 3x + y^2 + y = 22.5$. Then complete the square to put the equation in standard form. To do so, first rewrite $x^2 - 3x + y^2 + y = 22.5$ as $(x^2 - 3x + 2.25) - 2.25 + (y^2 + y + 0.25) - 0.25 = 22.5$. Second, add 2.25 and 0.25 to both sides of the equation: $(x^2 - 3x + 2.25) + (y^2 + y + 0.25) = 25$. Since $x^2 - 3x + 2.25 = (x - 1.5)^2$, $y^2 + y + 0.25 = (y + 0.5)^2$, and $25 = 5^2$, it follows that $(x - 1.5)^2 + (y + 0.5)^2 = 5^2$. Therefore, the radius of the circle is 5.

Choices B, C, and D are incorrect and may be the result of errors in manipulating the equation or of a misconception about the standard form of the equation of a circle in the xy -plane.

Q6**C****C.** 18

Choice C is correct. The perimeter of a triangle is the sum of the lengths of its sides. Since the given triangle is an isosceles right triangle, the length of each leg is the same and the length of the hypotenuse is equal to $\sqrt{2}$ times the length of a leg. Let x represent the length, in inches, of a leg of this isosceles right triangle. Therefore, the perimeter, in inches, of the triangle is $x + x + x\sqrt{2}$, or $2x + x\sqrt{2}$, which is equivalent to $x(2 + \sqrt{2})$. It's given that the perimeter of this triangle is $18 + 18\sqrt{2}$ inches. Thus, $x(2 + \sqrt{2}) = 18 + 18\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $x = \frac{18 + 18\sqrt{2}}{2 + \sqrt{2}}$. Multiplying the right-hand side of this equation by $\frac{2 - \sqrt{2}}{2 - \sqrt{2}}$ yields $x = \frac{36 + 36\sqrt{2} - 18\sqrt{2} - 36}{2}$, or $x = 9\sqrt{2}$. It follows that the length, in inches, of a leg of this isosceles right triangle is $9\sqrt{2}$. Therefore, the length, in inches, of the hypotenuse of this isosceles right triangle is $9\sqrt{2} \sqrt{2}$, or 18.

Choice A is incorrect. If this were the length of the hypotenuse, the perimeter would be $9 + 9\sqrt{2}$ inches.

Choice B is incorrect. This is the length, in inches, of a leg of this triangle, not the hypotenuse.

Choice D is incorrect. If this were the length of the hypotenuse, the perimeter would be $36 + 18\sqrt{2}$ inches.

The correct answer is 54. It's given that in triangle ABC , point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC . It follows that parallel segments DE and BC are intersected by sides AB and AC . If two parallel segments are intersected by a third segment, corresponding angles are congruent. Thus, corresponding angles C and AED are congruent and corresponding angles B and ADE are congruent. Since triangle ADE has two angles that are each congruent to an angle in triangle ABC , triangle ADE is similar to triangle ABC by the angle-angle similarity postulate, where side DE corresponds to side BC , and side AE corresponds to side AC . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{DE}{BC} = \frac{AE}{AC}$. Since point E lies on side AC , $AE + CE = AC$. It's given that $CE = 2AE$. Substituting $2AE$ for CE in the equation $AE + CE = AC$ yields $AE + 2AE = AC$, or $3AE = AC$. It's given that $BC = 162$. Substituting 162 for BC and $3AE$ for AC in the equation $\frac{DE}{BC} = \frac{AE}{AC}$ yields $\frac{DE}{162} = \frac{AE}{3AE}$, or $\frac{DE}{162} = \frac{1}{3}$. Multiplying both sides of this equation by 162 yields $DE = 54$. Thus, the length of line segment DE is 54.

Q8**24.5**Accept also: $49/2$

The correct answer is 24.5. It's given that a triangle is formed by connecting the three points shown, which are $-3, 4$, $5, 3$, and $4, -3$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $-3, 4$, $5, 4$, $5, -3$, and $-3, -3$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $5, -3$. This distance is $4 - -3$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of the rectangle is 78, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $5, 4$, and $5, 3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 4$ and $5, 3$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}81$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $4, -3$, $5, 3$, and $5, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 3$ and $5, -3$. This distance is $3 - -3$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, -3$ and $4, -3$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}16$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $-3, -3$, and $4, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $4, -3$ and $-3, -3$. This distance is $4 - -3$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $-3, 4$ and $-3, -3$. This distance is $4 - -3$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}77$, or 24.5. Thus, the area, in square units, of the triangle formed by connecting the three points shown is $56 - 4 - 3 - 24.5$, or 24.5. Note that 24.5 and $49/2$ are examples of ways to enter a correct answer.

Q9**16**

The correct answer is 16. An equation of a circle in the xy -plane can be written as $x - t^2 + y - u^2 = r^2$, where the center of the circle is t, u , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $x + 5^2 + y - 5^2 = 4$, which is equivalent to $x + 5^2 + y - 5^2 = 2^2$. Therefore, the center of circle A is $-5, 5$ and the radius of circle A is 2. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $-5, 5$ and the radius of circle B is 22, or 4. Substituting -5 for t , 5 for u , and 4 for r into the equation $x - t^2 + y - u^2 = r^2$ yields $x + 5^2 + y - 5^2 = 4^2$, which is equivalent to $x + 5^2 + y - 5^2 = 16$. It follows that the equation of circle B in the xy -plane is $x + 5^2 + y - 5^2 = 16$. Therefore, the value of k is 16.

Q10**.9811**

Accept also: 52/53

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is 53. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let x represent TS , the length of the other leg of the triangle shown. Therefore, $53^2 = (\sqrt{105})^2 + x^2$, or $2,809 = 105 + x^2$. Subtracting 105 from both sides of this equation yields $2,704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, TS , the length of the other leg of the triangle shown, is 52. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle R is 52, and the length of the hypotenuse is 53. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that $52/53$ or .9811 are examples of ways to enter a correct answer.

NotesAccept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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